

# Supplementary material

Psycholinguistic prediction of negative affect, positive affect, and life satisfaction on Weibo

## Scope and analysis chronology

The primary analysis estimates retrospective between-participant prediction of three stored questionnaire-derived outcomes. The supplied 1,427 rows represented 1,037 participants. Removing 195 exact duplicates left 1,232 distinct observations; selecting the numerically earliest questionnaire record per participant produced the 1,037-record primary sample. A single shuffled split assigned 933 participants to development and 104 to the holdout. All three requested outcomes and the supplied 120-feature representation were retained, with the input and participant-handling rules fixed before model comparison.

Primary model-selection rules, the two planned comparison types, record and feature sensitivities, and the development-only rule for selecting interpretation features were specified before inspection of the primary holdout performance. The separate exploratory extension was planned after that inspection. Its nested development evaluation repeats candidate selection within outer folds, but does not erase the effect of earlier information on the decision to conduct the extension. These two stages retain separate results throughout this supplement.

## Primary model development and evaluation

### Outcome units and selection score

NA, PA, and LS denote negative affect, positive affect, and life satisfaction. Although the retained outcomes range from 0 to 1, the transformation from administered questionnaire responses is unverified. All final RMSEs and feature attributions use the supplied stored units. Fold-local target standardization is an internal modeling operation, not an inferred reconstruction of the administered scoring rule.

For fold  $k$  and outcome  $j$ , let  $s_{jk}$  be the population standard deviation of that outcome in the fold's training participants. Selection used

$$Q = \frac{1}{5} \sum_{k=1}^5 \frac{1}{3} \sum_{j=1}^3 \frac{\text{RMSE}_{jk}}{s_{jk}}.$$

Independent families selected each outcome's hyperparameters from its own mean normalized error, after which the assembled family was compared on the macro score. Shared families selected a common setting. Ties followed the listed candidate order. These development scores guide selection and are not unbiased estimates of a procedure that has already been tuned on the same folds.

Each fold fitted median imputation, min-max input scaling, and target standardization using only its training participants. No feature selection, outcome-based record selection, value clipping, or outlier deletion was used in the primary analysis. The full primary holdout comparison is provided in Table S1; the grid and locked parameters are provided in Table S2.

**Table S1:** All primary model families on the holdout

| Model           | Outcome | $r$ [95% CI]             | $p$     | RMSE [95% CI]           | $R^2$  | $\rho$ |
|-----------------|---------|--------------------------|---------|-------------------------|--------|--------|
| Shared RF       | NA      | 0.247<br>[0.057, 0.419]  | 0.012   | 0.166<br>[0.146, 0.186] | 0.020  | 0.275  |
| Shared RF       | PA      | 0.284<br>[0.097, 0.452]  | 0.003   | 0.165<br>[0.143, 0.187] | 0.040  | 0.251  |
| Shared RF       | LS      | 0.152<br>[−0.042, 0.335] | 0.124   | 0.202<br>[0.181, 0.222] | 0.015  | 0.167  |
| Independent RF  | NA      | 0.230<br>[0.039, 0.405]  | 0.019   | 0.165<br>[0.144, 0.185] | 0.031  | 0.224  |
| Independent RF  | PA      | 0.325<br>[0.141, 0.487]  | < 0.001 | 0.163<br>[0.141, 0.186] | 0.057  | 0.315  |
| Independent RF  | LS      | 0.184<br>[−0.008, 0.364] | 0.061   | 0.201<br>[0.180, 0.222] | 0.025  | 0.166  |
| Independent ET  | NA      | 0.276<br>[0.088, 0.445]  | 0.005   | 0.165<br>[0.145, 0.186] | 0.028  | 0.267  |
| Independent ET  | PA      | 0.275<br>[0.087, 0.444]  | 0.005   | 0.165<br>[0.142, 0.187] | 0.037  | 0.274  |
| Independent ET  | LS      | 0.225<br>[0.033, 0.400]  | 0.022   | 0.200<br>[0.179, 0.220] | 0.034  | 0.161  |
| Ridge           | NA      | 0.274<br>[0.086, 0.443]  | 0.005   | 0.169<br>[0.148, 0.189] | −0.012 | 0.235  |
| Ridge           | PA      | 0.330<br>[0.147, 0.492]  | < 0.001 | 0.163<br>[0.140, 0.186] | 0.061  | 0.395  |
| Ridge           | LS      | 0.166<br>[−0.027, 0.348] | 0.092   | 0.202<br>[0.181, 0.222] | 0.021  | 0.161  |
| Multitask EN    | NA      | 0.331<br>[0.148, 0.493]  | < 0.001 | 0.166<br>[0.146, 0.186] | 0.024  | 0.344  |
| Multitask EN    | PA      | 0.402<br>[0.227, 0.552]  | < 0.001 | 0.165<br>[0.143, 0.187] | 0.038  | 0.408  |
| Multitask EN    | LS      | 0.168<br>[−0.025, 0.350] | 0.087   | 0.202<br>[0.182, 0.223] | 0.014  | 0.205  |
| Shared ET       | NA      | 0.312<br>[0.127, 0.476]  | 0.001   | 0.166<br>[0.145, 0.186] | 0.024  | 0.307  |
| Shared ET       | PA      | 0.306<br>[0.120, 0.471]  | 0.002   | 0.166<br>[0.142, 0.188] | 0.033  | 0.302  |
| Shared ET       | LS      | 0.227<br>[0.036, 0.402]  | 0.021   | 0.201<br>[0.180, 0.222] | 0.026  | 0.182  |
| Training mean   | NA      | Undefined                | —       | 0.169<br>[0.150, 0.190] | −0.021 | —      |
| Training mean   | PA      | Undefined                | —       | 0.170<br>[0.146, 0.192] | −0.016 | —      |
| Training mean   | LS      | Undefined                | —       | 0.205<br>[0.184, 0.225] | −0.009 | —      |
| Independent HGB | NA      | 0.075<br>[−0.119, 0.264] | 0.447   | 0.172<br>[0.150, 0.194] | −0.056 | 0.066  |
| Independent HGB | PA      | 0.219<br>[0.028, 0.395]  | 0.025   | 0.167<br>[0.146, 0.187] | 0.018  | 0.151  |
| Independent HGB | LS      | 0.190<br>[−0.003, 0.369] | 0.054   | 0.204<br>[0.180, 0.227] | 0.001  | 0.199  |

NA = negative affect; PA = positive affect; LS = life satisfaction. RF = random forest; ET = extra trees; EN = elastic net; HGB = histogram gradient boosting.  $n = 104$  in every row.  $p$  is the unadjusted two-sided Pearson test;  $\rho$  is Spearman correlation. Only the selected shared RF correlations were included in the three-test Holm family. Models are ordered by development score, not holdout correlation. RMSE retains stored units. Fisher- $z$  and 5,000-bootstrap intervals are reported.

**Table S2:** Primary grid and locked model settings

| Family          | Configurations                             | Selected settings                               | CV score |
|-----------------|--------------------------------------------|-------------------------------------------------|----------|
| Shared RF       | 6: leaf = 1, 5, 15; features = 0.5, 1.0    | leaf = 15; features = 1.0                       | 0.9963   |
| Independent RF  | 6: leaf = 1, 5, 15; features = 0.5, 1.0    | leaf = 15 / 15 / 15; features = 1.0 / 0.5 / 1.0 | 0.9965   |
| Independent ET  | 6: leaf = 1, 5, 15; features = 0.5, 1.0    | leaf = 15 / 15 / 15; features = 0.5 / 1.0 / 1.0 | 0.9968   |
| Ridge           | 5: alpha = 0.1, 1, 10, 100, 1000           | alpha = 1000 / 10 / 10                          | 0.9969   |
| Multitask EN    | 6: alpha = 0.001, 0.01, 0.1; L1 = 0.2, 0.8 | alpha = 0.01; L1 = 0.8                          | 0.9972   |
| Shared ET       | 6: leaf = 1, 5, 15; features = 0.5, 1.0    | leaf = 15; features = 0.5                       | 0.9977   |
| Training mean   | 1: training mean                           | None                                            | 1.0016   |
| Independent HGB | 6: nodes = 15, 31; L2 = 0, 1, 10           | nodes = 15 / 31 / 31; L2 = 10 / 1 / 10          | 1.0226   |

NA = negative affect; PA = positive affect; LS = life satisfaction. Slash-separated settings follow NA / PA / LS. The grid contains 42 configurations. Leaf denotes minimum samples per leaf; features is the fraction of input features considered at a split. All forests have 256 trees. HGB: 150 iterations, learning rate 0.05, minimum leaf size 20, no early stopping. EN: 10,000 maximum iterations, tolerance  $10^{-5}$ . CV score is macro normalized RMSE, not stored-unit holdout RMSE.

### Fixed-model uncertainty and comparison direction

RMSE is the square root of the mean squared difference between prediction and observation.  $R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y}_{\text{holdout}})^2$  uses the holdout mean in its denominator. It is not obtained by simply squaring the reported correlation. The training-mean baseline can consequently have negative  $R^2$ , and its constant predictions have undefined correlation.

The primary selected-model Pearson tests form a three-test Holm family. The paired prediction-error tests form a separate six-test family: selected model versus training mean and best shared procedure versus best independent ensemble, each for three outcomes. Every paired difference is first model minus second model. Pointwise RMSE-difference intervals use 5,000 paired participant resamples. Within-participant prediction swaps use 10,000 randomizations of the MSE-difference statistic and the plus-one correction. No interval is advertised as a simultaneous interval, and no nonsignificant test is interpreted as evidence of equivalence. Resampling fixes the learned models and partition; it does not repeat the complete model-development process.

### Planned handling sensitivities

The latest-record sensitivity retained the numerically last distinct observation for each participant. The singleton sensitivity retained participants with only one observation and therefore reduced the holdout to 85 participants. The feature sensitivity removed ind and the 13 unmapped codes, retaining 106 predictors. All retained participants kept their original development/holdout assignment, and model parameters remained those selected in the primary analysis. Table S3 reports the selected shared forest in all three sensitivities alongside the primary result. Full all-family sensitivity outputs remain available in the source repository; no new choice among these alternatives was used to replace the primary analysis.

**Table S3:** Selected shared RF under planned sensitivities

| Scenario            | <i>n</i> | Outcome | <i>r</i> [95% CI]        | RMSE [95% CI]           | <i>R</i> <sup>2</sup> |
|---------------------|----------|---------|--------------------------|-------------------------|-----------------------|
| Primary             | 104      | NA      | 0.247<br>[0.057, 0.419]  | 0.166<br>[0.146, 0.186] | 0.020                 |
| Primary             | 104      | PA      | 0.284<br>[0.097, 0.452]  | 0.165<br>[0.143, 0.187] | 0.040                 |
| Primary             | 104      | LS      | 0.152<br>[−0.042, 0.335] | 0.202<br>[0.181, 0.222] | 0.015                 |
| Latest record       | 104      | NA      | 0.259<br>[0.070, 0.431]  | 0.174<br>[0.154, 0.194] | 0.024                 |
| Latest record       | 104      | PA      | 0.286<br>[0.099, 0.454]  | 0.163<br>[0.144, 0.182] | 0.048                 |
| Latest record       | 104      | LS      | 0.101<br>[−0.094, 0.288] | 0.207<br>[0.185, 0.228] | 0.008                 |
| Singleton IDs       | 85       | NA      | 0.303<br>[0.097, 0.485]  | 0.171<br>[0.147, 0.195] | 0.025                 |
| Singleton IDs       | 85       | PA      | 0.301<br>[0.094, 0.484]  | 0.161<br>[0.138, 0.184] | 0.055                 |
| Singleton IDs       | 85       | LS      | 0.173<br>[−0.042, 0.372] | 0.207<br>[0.185, 0.229] | 0.027                 |
| Restricted features | 104      | NA      | 0.301<br>[0.115, 0.466]  | 0.165<br>[0.146, 0.185] | 0.031                 |
| Restricted features | 104      | PA      | 0.268<br>[0.080, 0.438]  | 0.165<br>[0.142, 0.188] | 0.035                 |
| Restricted features | 104      | LS      | 0.125<br>[−0.069, 0.310] | 0.203<br>[0.182, 0.223] | 0.008                 |

NA = negative affect; PA = positive affect; LS = life satisfaction. Each scenario refits the frozen primary settings without re-tuning, retaining participant assignments. Restricted features removes ind plus A, B, C1–C10, and D, leaving 106 predictors. Values are holdout results; intervals are pointwise. Complete all-family sensitivity results remain in the frozen repository results/primary/metrics.csv.

Three repetitions of five-fold development CV used seeds 20261023, 20261024, and 20261025 with frozen primary configurations. Table S4 preserves the reported precision of the baseline supplementary summaries. Because these participants had informed parameter selection, the repetition assesses behavior under different folds rather than supplying an independently selected validation estimate.

**Table S4:** Repeated development cross-validation under frozen settings

| Model          | Outcome | Mean normalized RMSE | Across-fold SD |
|----------------|---------|----------------------|----------------|
| Shared RF      | NA      | 0.9993               | 0.0627         |
| Shared RF      | PA      | 0.9995               | 0.0523         |
| Shared RF      | LS      | 0.9923               | 0.0578         |
| Independent RF | NA      | 1.0034               | 0.0647         |
| Independent RF | PA      | 1.0003               | 0.0548         |
| Independent RF | LS      | 0.9899               | 0.0615         |
| Training mean  | NA      | 1.0005               | 0.0647         |
| Training mean  | PA      | 1.0011               | 0.0568         |
| Training mean  | LS      | 1.0007               | 0.0570         |

NA = negative affect; PA = positive affect; LS = life satisfaction. Three repetitions of five-fold CV; seeds 20261023, 20261024, 20261025. Values are reproduced to the precision reported in the baseline supplement. Development observations already informed tuning. SD is descriptive fold variation, not a sampling confidence interval or an independent nested estimate.

## Attribution and perturbation diagnostics

TreeSHAP used the explicitly specified tree-path-dependent background on the development participants. Local contributions were summarized as the mean absolute contribution for each feature and outcome, in stored units (Lundberg et al., 2020). The ten highest-ranked features per outcome (Table S5) formed a union of 14 features. No interpretation-feature selection used holdout error.

**Table S5:** Top ten development-set TreeSHAP features per outcome

| Rank | NA feature       | SHAP   | PA feature       | SHAP   | LS feature       | SHAP   |
|------|------------------|--------|------------------|--------|------------------|--------|
| 1    | Sexual           | 0.0041 | Sexual           | 0.0045 | Family           | 0.0061 |
| 2    | C10 <sup>†</sup> | 0.0040 | We               | 0.0039 | We               | 0.0056 |
| 3    | C2 <sup>†</sup>  | 0.0036 | Family           | 0.0030 | C10 <sup>†</sup> | 0.0052 |
| 4    | D <sup>†</sup>   | 0.0027 | D <sup>†</sup>   | 0.0026 | Sexual           | 0.0044 |
| 5    | Angery           | 0.0025 | Angery           | 0.0023 | D <sup>†</sup>   | 0.0039 |
| 6    | Period           | 0.0021 | C10 <sup>†</sup> | 0.0020 | Angery           | 0.0036 |
| 7    | We               | 0.0020 | SheHe            | 0.0019 | C2 <sup>†</sup>  | 0.0032 |
| 8    | SheHe            | 0.0019 | C2 <sup>†</sup>  | 0.0017 | Quote            | 0.0027 |
| 9    | Family           | 0.0017 | Hate             | 0.0016 | Period           | 0.0027 |
| 10   | tFuture          | 0.0016 | tNow             | 0.0016 | HarmVirtue       | 0.0025 |

NA = negative affect; PA = positive affect; LS = life satisfaction. Mean absolute contributions, stored outcome units; shared RF,  $n = 933$ . <sup>†</sup> Unmapped dictionary code. The additive reconstruction error recorded in the analysis was  $2.22 \times 10^{-16}$ . Absolute magnitudes do not supply direction.

Each selected feature was shuffled independently across the 104 holdout participants 20 times. Table S6 reports the mean, SD, minimum, and maximum RMSE change across those shuffles. Negative change means the fitted model’s error decreased after that perturbation; it is not evidence that increasing the feature would improve well-being. Correlated or overlapping categories may substitute for each other, while perturbation may alter their usual combinations. These diagnostics concern a fitted input–output mapping. A change to that mapping’s inputs is distinct from an intervention on a person’s experiences or behavior (Janzing et al., 2020).

**Table S6:** Holdout permutation diagnostics for the 14 development-selected features

| Outcome | Feature    | Mean      | SD       | Minimum   | Maximum  |
|---------|------------|-----------|----------|-----------|----------|
| NA      | Angery     | 0.000106  | 0.000290 | −0.000426 | 0.000614 |
| NA      | C10        | 0.000320  | 0.000501 | −0.000810 | 0.001203 |
| NA      | C2         | 0.000249  | 0.000435 | −0.000507 | 0.000955 |
| NA      | D          | 0.000074  | 0.000376 | −0.000520 | 0.000837 |
| NA      | Family     | −0.000084 | 0.000180 | −0.000416 | 0.000186 |
| NA      | HarmVirtue | 0.000154  | 0.000120 | −0.000108 | 0.000444 |
| NA      | Hate       | 0.000195  | 0.000131 | −0.000065 | 0.000396 |
| NA      | Period     | 0.000138  | 0.000257 | −0.000338 | 0.000574 |
| NA      | Quote      | 0.000009  | 0.000214 | −0.000559 | 0.000355 |
| NA      | Sexual     | 0.000081  | 0.000522 | −0.000992 | 0.000873 |
| NA      | SheHe      | 0.000256  | 0.000273 | −0.000207 | 0.000673 |
| NA      | We         | −0.000279 | 0.000359 | −0.000946 | 0.000324 |
| NA      | tFuture    | −0.000021 | 0.000170 | −0.000339 | 0.000366 |
| NA      | tNow       | −0.000092 | 0.000130 | −0.000374 | 0.000228 |
| PA      | Angery     | 0.000227  | 0.000296 | −0.000090 | 0.000833 |

Table S6 (continued)

| Outcome | Feature    | Mean      | SD       | Minimum   | Maximum   |
|---------|------------|-----------|----------|-----------|-----------|
| PA      | C10        | 0.000316  | 0.000180 | -0.000021 | 0.000614  |
| PA      | C2         | -0.000188 | 0.000333 | -0.000746 | 0.000356  |
| PA      | D          | 0.000224  | 0.000299 | -0.000309 | 0.000712  |
| PA      | Family     | -0.000085 | 0.000342 | -0.000967 | 0.000444  |
| PA      | HarmVirtue | 0.000321  | 0.000159 | 0.000048  | 0.000634  |
| PA      | Hate       | 0.000560  | 0.000257 | 0.000238  | 0.001218  |
| PA      | Period     | -0.000093 | 0.000271 | -0.000510 | 0.000505  |
| PA      | Quote      | 0.000227  | 0.000169 | -0.000069 | 0.000509  |
| PA      | Sexual     | 0.000002  | 0.000544 | -0.000886 | 0.000925  |
| PA      | SheHe      | 0.000095  | 0.000340 | -0.000687 | 0.000894  |
| PA      | We         | 0.000233  | 0.000415 | -0.000350 | 0.001027  |
| PA      | tFuture    | 0.000016  | 0.000129 | -0.000293 | 0.000247  |
| PA      | tNow       | 0.000306  | 0.000208 | -0.000082 | 0.000707  |
| LS      | Angery     | -0.000674 | 0.000388 | -0.001401 | 0.000003  |
| LS      | C10        | 0.000065  | 0.000628 | -0.001121 | 0.001287  |
| LS      | C2         | -0.000886 | 0.000448 | -0.001935 | -0.000297 |
| LS      | D          | 0.000928  | 0.000578 | 0.000023  | 0.001718  |
| LS      | Family     | 0.001030  | 0.000690 | -0.000330 | 0.002222  |
| LS      | HarmVirtue | 0.000088  | 0.000293 | -0.000456 | 0.000708  |
| LS      | Hate       | 0.000462  | 0.000248 | 0.000018  | 0.000987  |
| LS      | Period     | -0.000179 | 0.000255 | -0.000637 | 0.000188  |
| LS      | Quote      | 0.000881  | 0.000360 | 0.000221  | 0.001592  |
| LS      | Sexual     | -0.000169 | 0.000657 | -0.001564 | 0.000777  |
| LS      | SheHe      | 0.000203  | 0.000277 | -0.000309 | 0.000771  |
| LS      | We         | -0.000935 | 0.000742 | -0.002600 | 0.000295  |
| LS      | tFuture    | 0.000369  | 0.000215 | 0.000032  | 0.000694  |
| LS      | tNow       | 0.000369  | 0.000261 | -0.000178 | 0.000896  |

NA = negative affect; PA = positive affect; LS = life satisfaction.  $n = 104$ ; 20 shuffles per feature. All quantities describe change in RMSE (permuted minus unpermuted) in stored units. Positive changes denote worse prediction after permutation. SD and range describe shuffle variation, not sampling uncertainty. The feature set is the union of the top ten development SHAP features per outcome.

## Exploratory optimization after primary holdout inspection

### Fixed 22-configuration extension

The extension retained the original 933 development and 104 holdout participants, the earliest-record rule, and all 120 primary features. It compared the mean and original shared RF references; ridge regression with alpha 10, 100, or 1000 on raw or log1p-transformed features; radial-basis support-vector regression with  $C = 0.1, 1, \text{ or } 10$  and epsilon 0.1; shallow histogram boosting with 7 leaves, 100 iterations, learning rate 0.05, and L2 10 or 100; independent elastic net with alpha 0.01 or 0.1 and L1 mixing ratio 0.1 or 0.5; partial least squares with 2, 5, or 10 components; and independent forests with minimum leaf size 30 or 60, 256 trees, and a feature fraction of 0.5.

Median imputation and any input scaling or log transform occurred within training folds. Outcomes were standardized within fitting for all non-PLS models; PLS used its own training-only scaling. One candidate was selected jointly across outcomes using macro normalized RMSE, unlike the primary independent-family procedure that allowed outcome-specific parameter choices. The finite grid, tie rule, and stopping rule were

fixed before any new configuration was run. The specified 1% improvement heuristic refers to outer-fold normalized error, not clinical usefulness or statistical significance.

### Nested development evaluation and final candidate

Five outer folds each ran five inner folds, using inner seed 20260924 plus the outer-fold index. In each outer training sample, the 22-candidate selection was repeated and its selected candidate evaluated on that outer validation sample. The original shared RF and mean references were evaluated on the same outer observations. Nesting selection in this manner addresses overfitting of the selection criterion for the specified candidate set (Cawley & Talbot, 2010). Table S7 gives outcome-level outer-fold means and SDs; these SDs summarize the five folds and are not sampling confidence intervals.

**Table S7:** Exploratory outer-fold normalized prediction error

| Procedure          | Outcome | Mean   | Across-fold SD |
|--------------------|---------|--------|----------------|
| Mean baseline      | NA      | 1.0020 | 0.0463         |
| Mean baseline      | PA      | 1.0002 | 0.0836         |
| Mean baseline      | LS      | 1.0001 | 0.0605         |
| Original shared RF | NA      | 1.0027 | 0.0473         |
| Original shared RF | PA      | 0.9986 | 0.0857         |
| Original shared RF | LS      | 0.9991 | 0.0578         |
| Selected procedure | NA      | 1.0044 | 0.0481         |
| Selected procedure | PA      | 0.9997 | 0.0822         |
| Selected procedure | LS      | 0.9976 | 0.0572         |

NA = negative affect; PA = positive affect; LS = life satisfaction. Five outer folds; candidate selection repeats five-fold inner CV in each. Macro means: baseline 1.0007, original RF 1.0001, selected procedure 1.0006. These are normalized errors, not holdout RMSE in stored units. The entire extension is exploratory because earlier development and holdout results were known. SD is descriptive.

The macro outer-fold scores were 1.0007301 for the mean, 1.0001398 for the original shared forest, and 1.0005805 for the selected procedure. Relative improvement, defined as  $(Q_{\text{original}} - Q_{\text{selected}})/Q_{\text{original}}$ , was  $-0.00044063$ : the selected procedure had approximately 0.044% higher error. The 1% improvement heuristic was therefore not met. The full-development five-fold search selected an independent forest with leaf size 60 (Table S8); its tuning score is distinct from the nested performance of the selection procedure.

**Table S8:** Exploratory 22-configuration search on all development participants

| Candidate          | Setting    | CV normalized RMSE |
|--------------------|------------|--------------------|
| Mean               | Reference  | 1.0007             |
| Original shared RF | Reference  | 1.0001             |
| Ridge, raw         | alpha=10   | 1.0703             |
| Ridge, raw         | alpha=100  | 1.0298             |
| Ridge, raw         | alpha=1000 | 1.0013             |
| Ridge, log1p       | alpha=10   | 1.0709             |
| Ridge, log1p       | alpha=100  | 1.0301             |
| Ridge, log1p       | alpha=1000 | 1.0013             |
| RBF SVR            | C=0.1      | 1.0040             |
| RBF SVR            | C=1        | 1.0215             |

Table S8 (continued)

| Candidate      | Setting            | CV normalized RMSE |
|----------------|--------------------|--------------------|
| RBF SVR        | C=10               | 1.1019             |
| Shallow HGB    | L2=10              | 1.0139             |
| Shallow HGB    | L2=100             | 1.0013             |
| Elastic net    | alpha=0.01; L1=0.1 | 1.0654             |
| Elastic net    | alpha=0.01; L1=0.5 | 1.0446             |
| Elastic net    | alpha=0.1; L1=0.1  | 1.0173             |
| Elastic net    | alpha=0.1; L1=0.5  | 0.9984             |
| PLS            | components=2       | 1.0056             |
| PLS            | components=5       | 1.0271             |
| PLS            | components=10      | 1.0460             |
| Independent RF | leaf=30            | 0.9979             |
| Independent RF | leaf=60            | 0.9969             |

Development  $n = 933$ , five-fold selection. One candidate is chosen across all outcomes; these tuning scores are not independent performance estimates. RF leaf=60 was selected. Raw denotes untransformed inputs before fold-local scaling; log1p denotes a fold-local transformation. L1 is the elastic-net mixing ratio. Main-text nested estimates concern the selected procedure, not simply this winning configuration.

### Previously seen holdout

Only the final selected candidate was evaluated in the extension on the previously inspected holdout. Table S9 retains its correlations, errors, and intervals. Table S10 compares it with the training mean and original shared forest. An exploratory positive-affect comparison with the mean yielded Holm-adjusted  $p = 0.011$ , while every comparison with the original shared forest had adjusted  $p = 1.000$ . These tests remain descriptive post-selection evidence on a reused evaluation sample.

Table S9: Exploratory selected model on the previously seen holdout

| Outcome | $r$ [95% CI]             | $p$     | RMSE [95% CI]              | $R^2$ | $\rho$ |
|---------|--------------------------|---------|----------------------------|-------|--------|
| NA      | 0.245<br>[0.055, 0.417]  | 0.012   | 0.1661<br>[0.1461, 0.1871] | 0.020 | 0.229  |
| PA      | 0.332<br>[0.149, 0.493]  | < 0.001 | 0.1645<br>[0.1415, 0.1886] | 0.045 | 0.339  |
| LS      | 0.164<br>[−0.029, 0.346] | 0.095   | 0.2023<br>[0.1821, 0.2223] | 0.016 | 0.138  |

NA = negative affect; PA = positive affect; LS = life satisfaction.  $n = 104$ . Independent RF: 256 trees, minimum leaf size 60, max features 0.5.  $p$  is unadjusted Pearson  $p$ ;  $\rho$  is Spearman correlation. RMSE retains stored units. The holdout had already been inspected during the primary analysis; these are exploratory estimates, not an independent replication.

**Table S10:** Exploratory paired comparisons on the reused holdout

| Comparison              | Outcome | $\Delta$ RMSE | 95% CI             | $p$ (swap) | $p$ (Holm) |
|-------------------------|---------|---------------|--------------------|------------|------------|
| Candidate – mean        | NA      | –0.003408     | [–0.0067, –0.0004] | 0.034      | 0.168      |
| Candidate – mean        | PA      | –0.005163     | [–0.0082, –0.0021] | 0.002      | 0.011      |
| Candidate – mean        | LS      | –0.002525     | [–0.0064, 0.0014]  | 0.204      | 0.817      |
| Candidate – original RF | NA      | –0.000023     | [–0.0022, 0.0022]  | 0.986      | 1.000      |
| Candidate – original RF | PA      | –0.000396     | [–0.0025, 0.0016]  | 0.712      | 1.000      |
| Candidate – original RF | LS      | –0.000091     | [–0.0029, 0.0027]  | 0.948      | 1.000      |

NA = negative affect; PA = positive affect; LS = life satisfaction.  $n = 104$ . Differences are candidate minus comparator. Pointwise intervals: 5,000 paired bootstraps; tests: 10,000 MSE prediction swaps; Holm covers six rows. The legacy source field `shared_vs_independent` denotes candidate minus original RF in its last three rows; labels here follow the actual `model_a` and `model_b` fields. No direction or value was reversed. These tests do not renew confirmatory status.

### Source availability and unresolved measurement inputs

The publicly available project provides analysis code, configuration files, frozen aggregate outputs, and evidence notes. It does not include participant-level source records, original posts, individual predictions, or fitted model objects. These restricted inputs would be required to rerun the complete analysis from participant data. The independent writing package contains the aggregate sources used to typeset the present tables and figure, together with their provenance and checks. Table generation and bilingual typesetting are distinct from replication of model fitting.

Unverified score transformation, administered item wording and reporting period, dictionary versions and code mappings, and exact collection timing remain unresolved. Demographic subgroup validation, raw-text re-extraction, odd/even-text reliability, and item-level reliability could not be performed from the available aggregates. Recorded timestamps indicate that language features may include post-questionnaire text; participant separation alone does not create prospective ordering. These restrictions define the scope of the primary and exploratory estimates alike.

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